YouTrap: You're Trapped in the YouTube Filter Bubble

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Motivation 1

- YouTube's personalization narrows exposure and reinforces biased, polarized information spaces, which can lead to various personal and social problems.
- Users often experience filter bubbles but lack awareness of when and how they are trapped.
- Existing research rarely reflects real YouTube interactions or helps users diagnose their own bubbles.

→ **We need a system that helps users see and understand their personalized content bubble.**

Problem Statement 2

Users have no intuitive way to see how YouTube's algorithm shapes their content environment or to recognize when they are **trapped in a filter bubble**.

Our goal is to build a visualization system that makes personalized content bubbles visible and enables users to recognize their **biased information environment**.

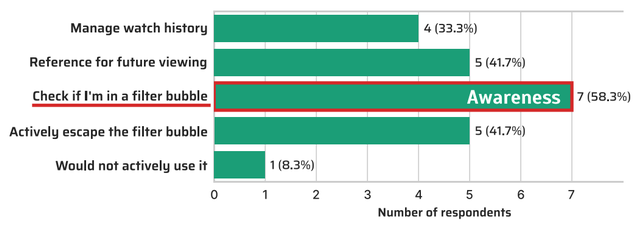
Pilot Study 3

- **Participants:** 12 users (avg. age 22.8).
- **Awareness of the "filter bubble" concept:** **No (91.7%)**, Yes (8.3%)
- **Experience with filter bubbles:** **83.3%** of participants reported having **experienced a filter bubble**.
- **Perceptions of filter bubbles:** **Negative (58.3%)**, Neutral (25%), Positive (16.7%)

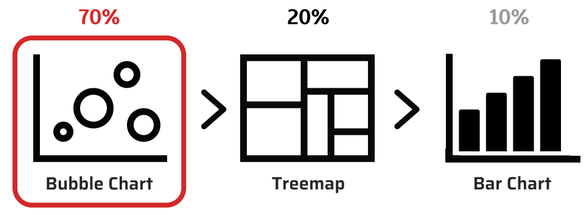
"Information can become biased." "Thinking patterns may become fixed." > "It's convenient to view only my interests."

- Which topics in the YouTube category list are likely to **generate biased opinions or controversy**? **People & Society (100%)**, **Law & Government (83.3%)**

〈 Intended Use of YouTrap 〉



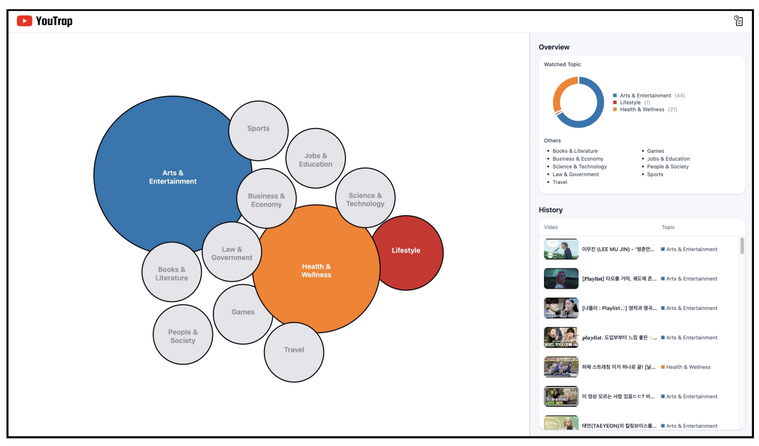
〈 Results on the chart type that best represents filter bubbles〉



The system's primary purpose is **awareness**, helping users recognize if and how they are trapped in filter bubbles.

System Design 4

[Main View] The bubble represents **a filter bubble**, with its border indicating a sense of being **"trapped"** inside it.



Topics

1. Arts & Entertainment
2. Books & Literature
3. Games
4. Lifestyle
5. Health & Wellness
6. Business & Economy
7. Jobs & Education
8. Science & Technology
9. People & Society
10. Law & Government
11. Sports
12. Travel

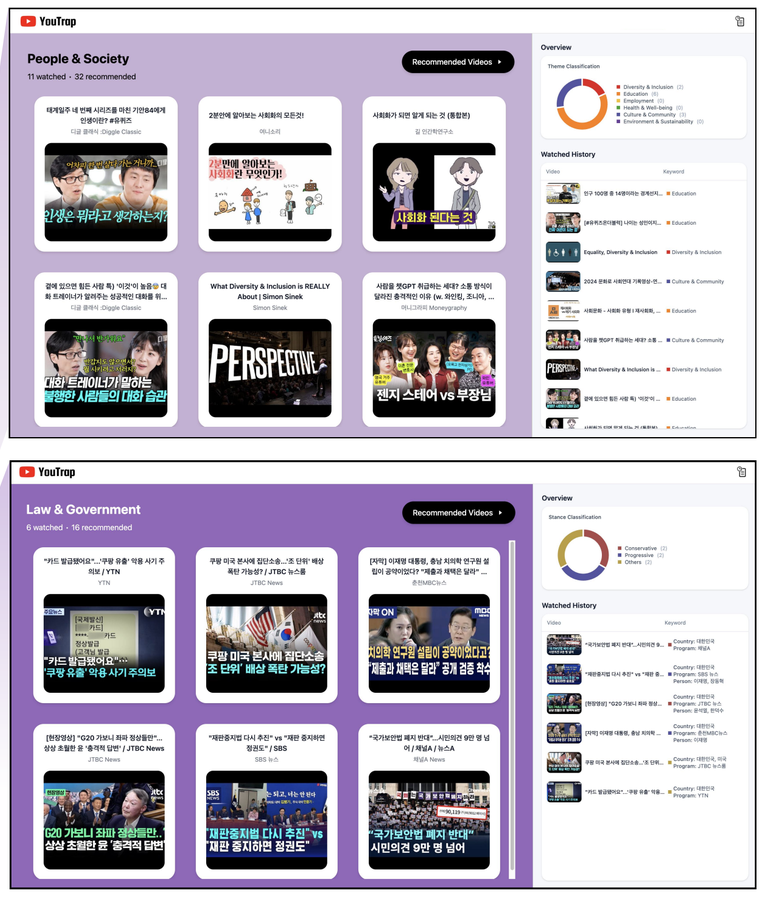
Additional Classification Layer for Sensitive Topics

Topic Classification
Videos are classified by the social issues they address, not by the stance they take.

Stance Classification
Tags are based on the video's topic, not the creator's stance.

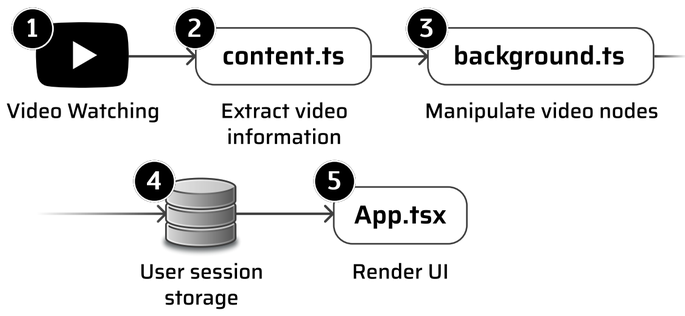
Keyword Extraction
Country
Program
Person

[Detail View] Selecting one takes you to the video link, and **"Recommended Videos"** displays a list of suggested videos.



Implementation 5

- React + TypeScript + D3
- Chrome Extension
- Session Storage — stores watched video data locally for privacy (serverless design)



[Extra Implementation for Various User Needs]

- **Long-term watch history to bubbles**

  Users can view bubbles of their long-term YouTube history by uploading their Google Takeout file.



Real-time **bubble** and **donut chart** updates based on accumulated **watch time** per topic.



- **Bubble visualization by watch time per each topic**

  Watch time accumulates per topic in real time, and users can toggle between **count** and **time** modes in the main view.

Evaluation 6

[Evaluation Method]

1. User Study

- **Participants:** 12 YouTube users (avg. age 28.3).
- **Goal:** To observe immediate awareness within **a short period of time**.
- **Apparatus:**
  The system ran as a Chrome extension on experimenters' laptops.
  All participants used their own logged-in YouTube accounts.
- **Procedure:**
  (1) Pre-survey and consent (~5 minutes)
  (2) Free video exploration on YouTube (~20 minutes)
  (3) System introduction and interaction with YouTrap (~10 minutes)
  (4) Post-survey (~10 minutes)

2. System Evaluation

- **Participant:** A frequent Arts & Entertainment and Sports viewer
- **Goal:** Evaluating whether **long-term watch history** produces accurate bubbles and supports filter-bubble awareness.
- **Procedure:** (1) Uploaded the watch history → (2) Selected 1 month → (3) Bubbles created

[Evaluation Results]

1. User Study Results

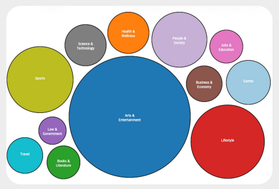
- **Qualitative**

Self-Awareness
(1) Awareness of Viewing Patterns
(2) Reflection on Unconscious Habits

External Influences
(1) Understanding Information Environment & Bubble Formation
(2) Understanding Recommendation Dynamics
(3) Understanding Context & Stance in Sensitive Domains

2. System Evaluation Results

- Since it's long-term history, bubbles form across various topics.
- But bubbles grow especially large in Arts & Entertainment and Sports, reflecting the participant's viewing pattern.



- **Quantitative**

(1) Distributions of Diversity Metrics and Algorithmic Divergence

Most users receive recommendations that closely follow their viewing patterns.

Restricted viewing diversity leads to restricted recommendation diversity, as supported by a strong positive correlation. **(r = 0.72, p = 0.012)**

(2) Survey items corresponding to SUS and NASA-TLX dimensions

High Usability
&
Low Cognitive Load

Contributions 7

[Individual-Level]

- Personal Bubble Awareness
- Self-Insight into Algorithmic Influence
- Better Control Over Personal Media Habits

[Societal-Level]

- Increased Visibility of Algorithmic Structures
- Support for a Diverse Information Ecosystem
- Foundation for Healthier Public Discourse

Limitations & Future Work 8

[Limitations]

- Lightweight model occasionally misclassifies video topics. ➡️ **Adopt a more powerful topic classification model.**
- Short-form contents are not included. ➡️ **Extend system to support the Shorts tab.**
- Short-duration study led to similar bubble sizes. ➡️ **Long-term studies to observe clearer bubble divergence.**

[Future Work]